17 ACIDS, BASES AND SALT PREPARATIONS

Acids were introduced in Chapter 16. In this chapter we will look more closely at some of the reactions of acids and how to make salts.



▲ Figure 17.1 Sodium chloride is common salt. You don't normally make it. It occurs naturally.

LEARNING OBJECTIVES

- Know the general rules for predicting the solubility of ionic compounds in water:
  - common sodium, potassium and ammonium compounds are soluble
  - all nitrates are soluble
  - common chlorides are soluble, except those of silver and lead(II)
  - common hydroxides are insoluble except for those of sodium, potassium and calcium (calcium hydroxide is slightly soluble)
- Understand acids and bases in terms of proton transfer
- Understand that an acid is a proton donor and a base is a proton acceptor
- Describe the reactions of hydrochloric acid, sulfuric acid and nitric acid with metals, bases and metal carbonates (excluding the reactions between nitric acid and metals) to form salts
- Know that metal oxides, metal hydroxides and ammonia can act as bases, and that alkalis are bases that are soluble in water
- Describe an experiment to prepare a pure, dry sample of a soluble salt, starting from an insoluble reactant
- Practical: Prepare a sample of pure, dry hydrated copper(II) sulfate crystals, starting from copper(II) oxide

CHEMISTRY ONLY

- Describe an experiment to prepare a pure, dry sample of a soluble salt, starting from an acid and alkali.
- Describe an experiment to prepare a pure, dry sample of an insoluble salt, starting from two soluble reactants.
- Practical: Prepare a sample of pure, dry lead(II) sulfate

SALTS

All acids contain hydrogen. When that hydrogen is replaced by a metal, the compound formed is called a **salt**. Magnesium sulfate is a salt, and so is zinc chloride, and so is potassium nitrate.

The formulae of some acids and salts are given in Table 17.1.

Table 17.1 Some acids and salts

Acid	Formula	Example of salt	Name of salts
hydrochloric acid	HCl	NaCl	chlorides
nitric acid	HNO_3	KNO_3	nitrates
sulfuric acid	H_2SO_4	$CuSO_4$	sulfates
ethanoic acid	CH_3COOH	CH_3COONa	ethanoates
phosphoric acid	H_3PO_4	K_3PO_4	phosphates

Sulfuric acid can be thought of as the **parent acid** of all the sulfates (the salts formed from sulfuric acid are all called sulfates). It doesn't matter if the replacement of the hydrogen can't be done directly. For example, you can't make copper(II) sulfate from copper and dilute sulfuric acid because they don't react. There are, however, other ways of making it from sulfuric acid – copper(II) sulfate is still a salt.

REMINDER

These salts contain the NH_4^+ ion.

Salts are also formed when the hydrogen in an acid is replaced with NH_4. These are ammonium salts, for example NH_4Cl (ammonium chloride) and $(NH_4)_2SO_4$ (ammonium sulfate).

REACTIONS OF ACIDS

Before we look at some practical methods to make salts, we need to study the reactions of acids.

REACTING ACIDS WITH METALS

KEY POINT

Of the common acids in the lab, nitric acid has much more complex reactions with metals. You won't be asked about this at International GCSE.

Simple dilute acids react with metals depending on their positions in the reactivity series.

- Metals below hydrogen in the series don't react with dilute acids.
- Metals above hydrogen in the series react to produce hydrogen gas.
- The higher the metal is in the reactivity series, the more vigorous the reaction. You would never mix metals such as sodium or potassium with acids because their reactions are too violent.

REMINDER

Remember: MASH

A summary equation for metals above hydrogen in the reactivity series:

metal + acid → salt + hydrogen

REACTIONS INVOLVING MAGNESIUM AND ACIDS

WITH DILUTE SULFURIC ACID

There is rapid fizzing and a colourless gas is evolved, which pops with a lighted splint (the test for hydrogen). The reaction mixture becomes very warm as heat is produced. The magnesium gradually disappears to leave a colourless solution of magnesium sulfate.

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

This is a displacement reaction. The more reactive magnesium has displaced the less reactive hydrogen.



▲ Figure 17.2 Magnesium reacting with dilute sulfuric acid

WITH DILUTE HYDROCHLORIC ACID

The reaction looks exactly the same. The only difference is that this time a solution of magnesium chloride is formed.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

EXTENSION WORK

Acids in solution form ions. Dilute sulfuric acid contains hydrogen ions and sulfate ions. Dilute hydrochloric acid contains hydrogen ions and chloride ions.

You can rewrite the equations above as ionic equations. In the case of sulfuric acid:

$$Mg(s) + 2H^+(aq) + SO_4^{2-}(aq) \rightarrow Mg^{2+}(aq) + SO_4^{2-}(aq) + H_2(g)$$

The sulfate ion hasn't been changed by the reaction. It is a spectator ion, and so we don't include it in the ionic equation:

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

Repeating this with hydrochloric acid, we find that the chloride ions are also spectator ions:

$$Mg(s) + 2H^+(aq) + 2Cl^-(aq) \rightarrow Mg^{2+}(aq) + 2Cl^-(aq) + H_2(g)$$

Not including the spectator ions produces the ionic equation:

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

The reactions look the same because they are the same. All solutions of acids contain hydrogen ions. That means that magnesium will react with any simple dilute acid in the same way.

REMINDER

Ionic compounds are present as separate ions in solution, so in $MgSO_4(aq)$ the $Mg^{2+}(aq)$ and $SO_4^{2-}(aq)$ ions are separated from each other.

REACTIONS INVOLVING ZINC AND ACIDS

Again, the reactions between zinc and the two acids look exactly the same. The reactions are slower because zinc is lower down the reactivity series than magnesium.

The equations are:

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$
$$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

EXTENSION WORK

The ionic equations are both the same:

$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

BASES

We saw in Chapter 16 that bases are substances that neutralise acids by combining with the hydrogen ions in them to produce water. When we are referring to a base at this level, we usually mean a metal oxide, a metal hydroxide or ammonia.

REACTING ACIDS WITH BASES

Metal oxides, such as copper(II) oxide and magnesium oxide, are bases.

REACTING DILUTE SULFURIC ACID WITH COPPER(II) OXIDE

The copper(II) oxide (black powder) reacts with hot dilute sulfuric acid to produce a blue solution of copper(II) sulfate.

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

All the metal oxide and acid combinations that you will meet at International GCSE behave in exactly the same way as the reaction between copper(II) oxide and dilute sulfuric acid, that is, they produce a salt and water. Most need to be heated for the reaction to start.

Copper(II) oxide is an ionic compound containing the O^{2-} ion. What has happened in this reaction is that the H^+ ions from the acid have combined with the O^{2-} ions to form water (H_2O).

The general equation for the reaction of a metal oxide (base) with an acid is

metal oxide + acid → salt + water

This is a neutralisation reaction – the base neutralises the acid (water is formed).

BASES AND ALKALIS

Some metal oxides are soluble in water and react with it to form solutions of metal hydroxides, for example:

$$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$$

All the Group 1 oxides do this reaction, so for potassium oxide we would get:

$$K_2O(s) + H_2O(l) \rightarrow 2KOH(aq)$$

Most other metal oxides are not soluble in water.

Calcium oxide dissolves slightly to form calcium hydroxide:

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$$

Another alkali you will meet is a solution of ammonia (NH_3). The ammonia reacts with the water to form ammonium ions and hydroxide ions:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

All these solutions contain hydroxide ions. We learned in the last chapter that we can define alkalis as solutions that are a source of hydroxide (OH^-) ions, so all of these solutions are alkalis.

REACTING ACIDS WITH METAL HYDROXIDES

All metal hydroxides react with acids in a neutralisation reaction:

metal hydroxide + acid → salt + water

REACTING DILUTE HYDROCHLORIC ACID WITH SODIUM HYDROXIDE SOLUTION

Mixing sodium hydroxide solution and dilute hydrochloric acid produces a colourless solution so not much seems to have happened. But if you repeat the reaction with a thermometer in the beaker, the temperature rises several degrees, showing that there has been a chemical change. Sodium chloride solution has been formed:

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

We saw in Chapter 16 that the ionic equation for this reaction is:

$$OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$$

All neutralisation reactions for an acid reacting with an alkali have the same ionic equation: they all involve the hydroxide ions from the alkali reacting with the H^+ ions from the acid to form water.

REACTING ACIDS WITH CARBONATES

Carbonates react with cold dilute acids to produce carbon dioxide gas.

A summary equation for acids and carbonates:

carbonate + acid → salt + carbon dioxide + water

THE REACTION BETWEEN COPPER(II) CARBONATE AND DILUTE ACIDS

Green copper(II) carbonate reacts with the common dilute acids to give a blue or blue-green solution of copper(II) sulfate, copper(II) nitrate or copper(II) chloride. Carbon dioxide gas is given off. You can recognise this because it turns limewater milky.

All the equations have the same form:

$$CuCO_3(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + CO_2(g) + H_2O(l)$$

$$CuCO_3(s) + 2HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + CO_2(g) + H_2O(l)$$

$$CuCO_3(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + CO_2(g) + H_2O(l)$$



▲ Figure 17.3 The reaction between copper(II) carbonate and dilute sulfuric acid.

THE REACTION BETWEEN SODIUM CARBONATE AND DILUTE ACIDS

Sodium carbonate is soluble in water. The equation for the reaction of sodium carbonate with hydrochloric acid is:

$$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

HINT

Carbonates are not mentioned as bases on the syllabus; only metal oxides, metal hydroxides and ammonia.

EXTENSION WORK

If we show the ions (remembering that HCl splits apart into its ions in solution):

$$2Na^+(aq) + CO_3^{2-}(aq) + 2H^+(aq) + 2Cl^-(aq) \rightarrow 2Na^+(aq) + 2Cl^-(aq) + CO_2(g) + H_2O(l)$$

By not including the spectator ions we obtain the ionic equation:

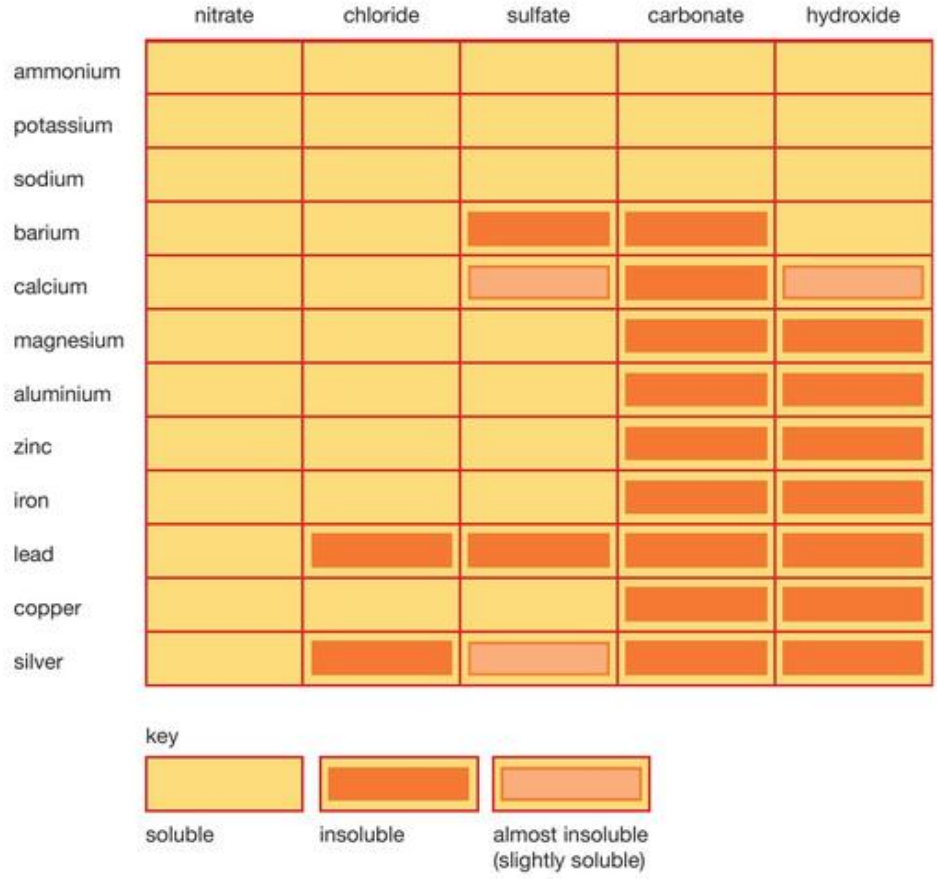
$$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$$

Because carbonate ions react by accepting hydrogen ions, we can classify carbonate ions as bases.

SALT PREPARATIONS

In the next sections we will learn about some practical methods to make salts. Before we can do this, we need to know which substances are soluble and which are insoluble in water because the method we use depends on this.

THE SOLUBILITY OF IONIC COMPOUNDS IN WATER



▲ Figure 17.4 **Solubility patterns**

NOTES ON THE TABLE

To keep the table simple, it includes one or two compounds (aluminium carbonate, for example) which don't actually exist. Don't worry about these. The problem won't arise at International GCSE.

There is no clear line between 'insoluble' and 'almost insoluble' compounds. The ones picked out as 'almost insoluble' include the more common ones that you might need to know about elsewhere in the course.

HINT

It can seem a bit frightening to have to remember all this, but it isn't as difficult as it looks at first sight.

Except for the carbonates and hydroxides, most of these compounds are soluble. Learn the exceptions in the sulfates and chlorides.

The reason for the exceptions in the carbonates and hydroxides is that all sodium, potassium and ammonium compounds are soluble.

Note that:

- all sodium, potassium and ammonium compounds are *soluble*
- all nitrates are *soluble*
- most common chlorides are *soluble*, except lead(II) chloride and silver chloride
- most common sulfates are *soluble*, except lead(II) sulfate, barium sulfate, silver sulfate and calcium sulfate
- most common carbonates are *insoluble*, except sodium, potassium and ammonium carbonates
- most metal hydroxides are *insoluble* (or *almost insoluble*), except sodium, potassium and ammonium hydroxides. Calcium hydroxide is slightly soluble in water.

MAKING SOLUBLE SALTS (EXCEPT SODIUM, POTASSIUM AND AMMONIUM SALTS)

These all involve reacting a solid with an acid. You can use any of the following mixtures:

- acid + metal (but only for the moderately reactive metals from magnesium to iron in the reactivity series)
- acid + metal oxide or hydroxide
- acid + carbonate.

Whatever mixture you use, the method is basically the same.

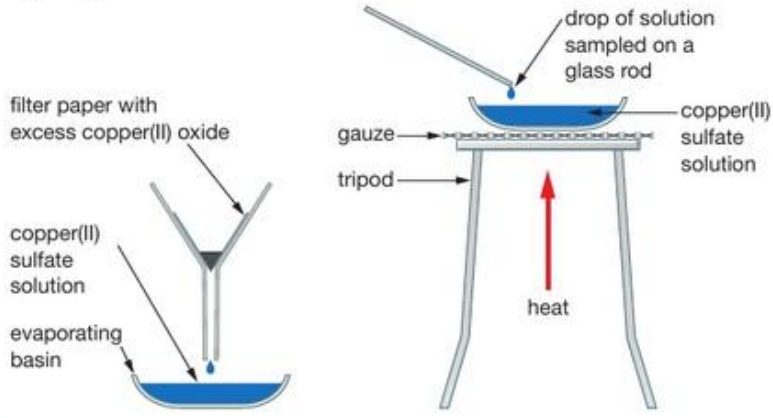
ACTIVITY 5

▼ PRACTICAL: MAKING COPPER(II) SULFATE CRYSTALS

> ! Safety Note: Direct heating of the acid is not recommended. Wear eye protection and do not evaporate the solution to dryness. The acid is best heated in a boiling tube placed in a water bath. Avoid contact with the crystals as they are harmful and irritating to both the skin and eyes.

The practical procedure for this is as follows:

- Measure 50 cm^3 of dilute sulfuric acid into a beaker and heat it on a tripod and gauze using a Bunsen burner.
- Add a spatula full of black copper(II) oxide and continue heating. If all the copper(II) oxide disappears add more copper(II) oxide until there is some left in the beaker. Stir the mixture well to make sure that no more will react. At this stage, we have added *excess* copper(II) oxide – there is more than enough to react with all the acid present. When there is copper(II) oxide left, we know that all the acid has been neutralised.
- Filter off the excess copper(II) oxide and transfer the filtrate (solution), which is blue, to an evaporating basin. The solution we have now is copper(II) sulfate.



▲ Figure 17.5 Making copper(II) sulfate crystals

The equation for the formation of the solution is:

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

- Heat the solution of copper(II) sulfate over a Bunsen burner to boil off some of the water and concentrate the solution.
- Keep heating until a saturated solution is formed. We can test this by dipping a glass rod into the solution. If crystals form on the glass rod when we remove it we know that the solution is very close to saturated and crystals will also begin to form in the solution.

- Stop heating the reaction mixture and allow it to cool slowly at room temperature so that larger crystals can form.
- Remove the blue crystals from the reaction mixture by filtration, or by just pouring off the remaining solution.
- The crystals can be dried by blotting them with a paper towel, or they can be left to dry in a warm place.

WHY NOT SIMPLY EVAPORATE THE SOLUTION TO DRYNESS?

It would seem much easier to just evaporate off all the water by boiling rather than crystallising the solution slowly, but evaporating to dryness wouldn't give you blue copper(II) sulfate crystals. Instead, you would produce a white powder of anhydrous copper(II) sulfate.

When many salts form their crystals water from the solution becomes chemically bound up with the salt. This is called *water of crystallisation*. A salt which contains water of crystallisation is said to be *hydrated*. We can show the crystallisation part of the reaction as:

$$CuSO_4(aq) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(s)$$

KEY POINT

'Anhydrous' means 'without water'.

!

Safety Note: Wear eye protection. Heating must be very gentle to avoid 'spitting' of hot particles.



▲ Figure 17.6 Copper(II) sulfate crystals

MAKING MAGNESIUM SULFATE CRYSTALS

We can add excess magnesium to sulfuric acid. This time the acid does not have to be heated.

When we add the magnesium the reaction mixture will fizz (hydrogen is given off). We keep adding magnesium until the fizzing stops and there is magnesium left in the beaker. This means that all the acid has reacted. The equation for the formation of the solution of magnesium sulfate is:

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

Again the solution is concentrated by heating it and allowed to crystallise. The crystallisation reaction is:

$$MgSO_4(aq) + 7H_2O(l) \rightarrow MgSO_4 \cdot 7H_2O(s)$$



▲ Figure 17.7 Dilute sulfuric acid with magnesium

HOW DO YOU KNOW WHETHER YOU NEED TO HEAT THE MIXTURE?

Carbonates react with dilute acids in the cold, and so does magnesium. Most other things that you are likely to come across need to be heated.



▲ Figure 17.8 Do you need to heat the mixture?

CHEMISTRY ONLY

MAKING SODIUM, POTASSIUM AND AMMONIUM SALTS

THE NEED FOR A DIFFERENT METHOD

In the method we've just been looking at, you add an excess of a solid to an acid, and then filter off the unreacted solid. You do this to make sure all the acid is used up. This method does not work for making sodium, potassium and ammonium salts.

The problem is that all sodium, potassium and ammonium compounds are soluble in water. The solid you added to the acid would not only react with the acid, but any excess would just dissolve in the water present. You wouldn't have any visible excess to filter off. There's no simple way of seeing when you have added just enough of the solid to neutralise the acid.

SOLVING THE PROBLEM BY DOING A TITRATION

You normally make these salts from sodium or potassium hydroxide or ammonia solution, but you can also use the carbonates. Fortunately, the solutions of all these are alkaline. That means you can find out when you have a neutral solution by using an indicator.

The method of finding out exactly how much of two solutions you need for them to neutralise each other is called titration (see Chapter 16). You carry out a titration using an indicator to tell you when you have added exactly enough of the acid to neutralise the alkali. Having found out how much acid and alkali are needed, you can make a pure solution of the salt by mixing those same volumes again, but without the indicator.

MAKING SODIUM SULFATE CRYSTALS

KEY POINT

Methyl orange is red in acidic solutions and yellow in alkaline ones. However, when it is used as an indicator in titrations (see Chapter 16) involving adding an acid to an alkali, an orange colour is usually obtained, which indicates the endpoint. Orange is a mixture of the red and yellow forms.

- 25 cm^3 of sodium hydroxide solution is transferred to a conical flask using a pipette, and a few drops of methyl orange are added as the indicator.
- Dilute sulfuric acid is run in from the burette until the indicator just turns from yellow to orange.
- The volume of acid needed is noted, and the same volumes of acid and alkali are mixed together in a clean flask *without any indicator*.

The equation for the formation of the solution is:

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

- The solution is heated to evaporate off some of the water until a saturated solution is formed. It is then left to cool so that crystals form.
- The crystals are finally separated from any remaining solution by filtration (or the remaining solution can be poured away).
- The crystals are dried by patting them dry with a paper towel or by leaving them in a warm place.

The equation for the crystallisation process is:

$$Na_2SO_4(aq) + 10H_2O(l) \rightarrow Na_2SO_4 \cdot 10H_2O(s)$$

EXTENSION WORK

Methyl orange actually changes colour between pH 3.1 and pH 4.4, so below pH 3.1 methyl orange is red and above pH 4.4 it is yellow.

Between these pHs methyl orange will be a mixture of red and yellow, that is, orange. This means that it is actually yellow in a solution with pH 7. This technique does not, therefore, give us the exact neutralisation point, but because the pH changes very rapidly near the endpoint of a titration, if we do the titration carefully we will be able to find the neutralisation point to the nearest drop of acid, which is very good!



▶ Figure 17.9 **Titration using methyl orange indicator**

MAKING SODIUM CHLORIDE CRYSTALS

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

You would need to do the titration using dilute hydrochloric acid rather than dilute sulfuric acid. You would then follow the same procedure as for sodium sulfate.

HINT

Once you have re-mixed the acid and the alkali without the indicator, you could evaporate the sodium chloride solution to dryness rather than crystallising it slowly. Sodium chloride crystals don't contain any water of crystallisation, so you could save time by evaporating all the water in one go. The disadvantage is that you obtain either a powder or very tiny crystals. In the exam you will not necessarily know whether something contains water of crystallisation or not, so it is always better to concentrate the solution and let crystals form slowly.

MAKING AMMONIUM SULFATE CRYSTALS

$$2NH_3(aq) + H_2SO_4(aq) \rightarrow (NH_4)_2SO_4(aq)$$

Using ammonia solution rather than sodium hydroxide solution makes no difference to the method. Although simple ammonium salts don't have water of crystallisation, you would still crystallise them slowly rather than evaporating them to dryness. Heating dry ammonium salts tends to break them up.

MAKING INSOLUBLE SALTS

PRECIPITATION REACTIONS



▲ Figure 17.10 A precipitate of silver chloride

The basic procedure is to mix solutions of two soluble salts to form an insoluble salt and a solution of a soluble one. For instance, to make insoluble silver chloride we would mix together solutions of silver nitrate and sodium chloride.

$$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$$

This is called a **precipitation reaction**. A **precipitate** is a solid that is formed by a chemical reaction involving liquids or gases. A precipitation reaction is simply a reaction that produces a precipitate. In this reaction, a white precipitate of silver chloride is produced. A white precipitate is formed because silver chloride won't dissolve in water, and so it is seen as a fine white solid. Figure 17.10 shows the results of this reaction.

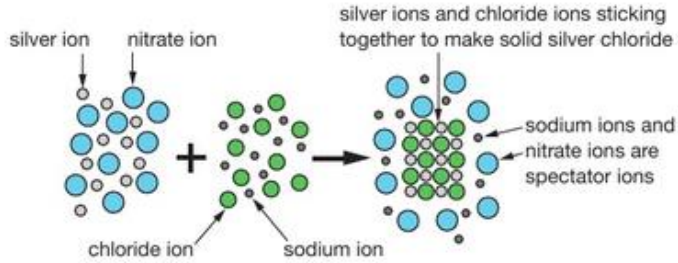
EXPLAINING WHAT'S HAPPENING

Silver nitrate solution contains silver ions and nitrate ions in solution. The positive and negative ions are attracted to each other, but the attractions aren't strong enough to make them stick together. Similarly, sodium chloride solution contains sodium ions and chloride ions; again, the attractions aren't strong enough for them to stick together.

When you mix the two solutions, the various ions meet each other. When silver ions meet chloride ions, the attractions are so strong that the ions clump together and form a solid. The sodium and nitrate ions remain in solution because they aren't sufficiently attracted to each other.

KEY POINT

The water molecules in the solutions have been left out to avoid cluttering the diagram.



▶ Figure 17.11 **Precipitation of silver chloride**

EXTENSION WORK

The ionic equation for a precipitation reaction is much easier to write than the full equation. All that is happening is that the ions of the insoluble salt are coming together to form the solid. The ionic equation simply shows that happening.

To work out the ionic equation for the precipitation reaction:

- Write down the formula for the precipitate/insoluble salt on the right-hand side of the equation.
- Write down the formulae for the ions that have come together to produce it on the left-hand side.
- Don't forget the state symbols.

So, for the formation of silver chloride we have:

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

or for the formation of barium sulfate:

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

You don't need to worry about the spectator ions because they aren't doing anything.



▲ Figure 17.12 A precipitate of lead(II) iodide

WHAT DO WE MIX TOGETHER TO MAKE INSOLUBLE SALTS?

Our procedure for making insoluble salts is to mix together two solutions containing soluble salts, but how do we know what to mix together? To determine this, two really useful facts are:

- all nitrates are soluble
- all sodium and potassium salts are soluble.

So all we have to do is mix the nitrate of the metal part of our insoluble salt with the sodium or potassium salt of the non-metal part.

For instance, if we have to make lead(II) iodide:

- the metal part is lead, so we use a solution of lead(II) nitrate
- the non-metal part is iodide, so we use a solution of sodium iodide.

The equation for the reaction is:

$$Pb(NO_3)_2(aq) + 2NaI(aq) \rightarrow PbI_2(s) + 2NaNO_3(aq)$$

Figure 17.12 shows the yellow precipitate of lead(II) iodide.

EXTENSION WORK

The ionic equation is:

$Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$

TO MAKE BARIUM SULFATE

We can use barium nitrate solution and potassium sulfate solution:

$$Ba(NO_3)_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KNO_3(aq)$$

The barium sulfate is a white precipitate.

There are other ways we could make barium sulfate. We could also use barium chloride, which is soluble in water. The sulfate part doesn't necessarily have to come from a salt; dilute sulfuric acid contains sulfate ions, so we can use that as well.

EXTENSION WORK

The ionic equation is:

$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

ACTIVITY 6

▼ PRACTICAL: PREPARING A PURE, DRY SAMPLE OF LEAD(II) SULFATE

! Safety Note: Wear eye protection. The lead nitrate and sulfate are toxic and all skin contact must be avoided.

The following procedure is used:

- Take 25 cm^3 of a solution of lead(II) nitrate in a beaker and add 25 cm^3 of a solution of sodium sulfate. The amounts don't really matter as we will get rid of any excess later.
- A white precipitate of lead(II) sulfate will form.
- The equation for the reaction is:

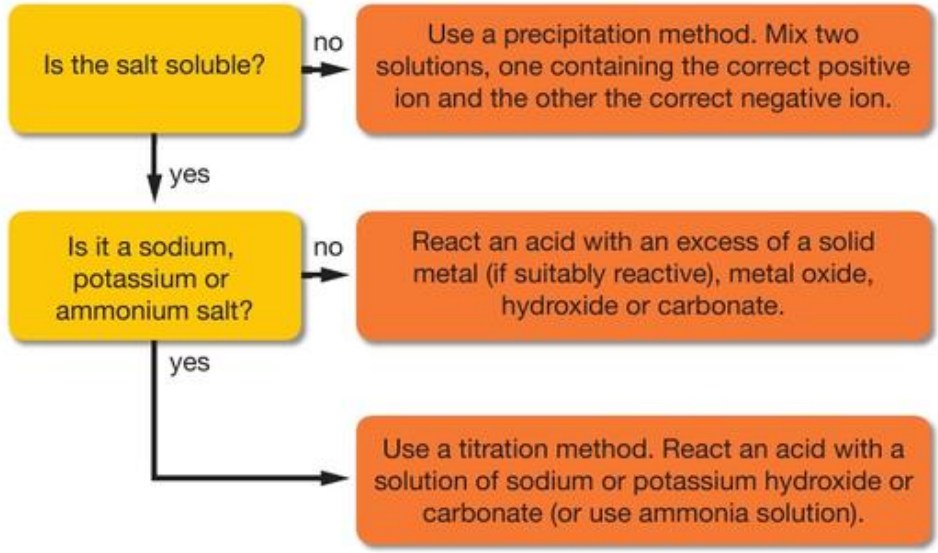
$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow PbSO_4(s) + 2NaNO_3(aq)$

- The reaction mixture is filtered.
- A white residue of lead(II) sulfate is left in the filter paper and a colourless solution of sodium nitrate and any excess reactants (either lead(II) nitrate or sodium sulfate) passes through into the beaker or conical flask. The white residue on the filter paper is the compound we want, but it is contaminated with solutions of sodium nitrate and the reactant that was in excess.

- Wash the residue with *distilled water* by pouring a 20 cm^3 portion (the amount is not important) into the filter paper and allowing it to filter through the paper. This washes away everything apart from the insoluble lead(II) sulfate. This should be repeated several times to make sure that our sample is not contaminated with anything.
- We then transfer the filter paper and lead(II) sulfate to a warm oven to dry (the water evaporates).

We must wash the lead(II) sulfate with distilled (pure) water because tap water contains dissolved substances. If we wash with tap water and then transfer our sample to the oven, the water will evaporate, leaving behind the dissolved solids from the water, which will contaminate our salt.

SUMMARISING THE METHODS OF MAKING SALTS



▲ Figure 17.13 Methods of making salts

END OF CHEMISTRY ONLY

THEORIES OF ACIDS AND BASES

In Chapter 16, we met the idea that an acid is something that produces $H^+(aq)$ ions in solution and an alkali is something that produces $OH^-(aq)$ ions in solution. This is called the Arrhenius theory of acids and bases. For example, when hydrogen chloride gas is dissolved in water to form hydrochloric acid the molecules dissociate to form H^+ ions:

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

A neutralisation reaction can be shown as H^+ from an acid reacting with OH^- from an alkali to form water.

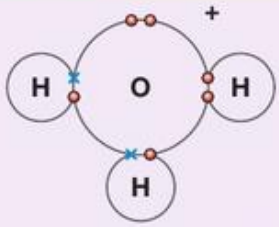
$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

The problem with this definition of acids and bases is that it only applies to reactions that occur in aqueous solution and is not more widely applicable.

HINT

You do not have to remember the names of the theories.

EXTENSION WORK



In the dot-and-cross diagram for the H_3O^+ ion you can see that the covalent bond that formed between the H_2O and the H^+ is unusual in that both the shared electrons come from the O. This is called a dative covalent bond or a co-ordinate covalent bond. Once it has been formed, it is exactly the same as a 'normal' covalent bond.

KEY POINT

You have already met the ammonium (NH_4^+) ion, which is obtained when H^+ is added to ammonia (NH_3). The hydroxonium ion is the equivalent ion for water. You do not have to remember the name *hydroxonium ion*.

KEY POINT

To understand the reaction between ammonia and hydrochloric acid in relation to the Arrhenius theory we have to realise that ammonia reacts with water to form OH^- ions:

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

and that hydrochloric acid dissociates to form H^+ ions:

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

and that the overall reaction involves H^+ ions reacting with OH^- ions to form water:

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

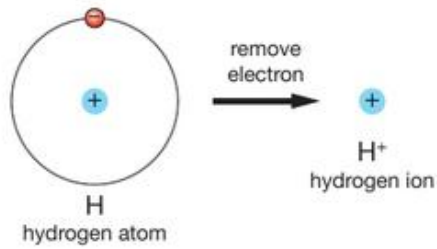


▲ Figure 17.17 Ammonia and hydrogen chloride gases also react to form ammonium chloride (the white smoke): $NH_3(g) + HCl(g) \rightleftharpoons NH_4^+Cl^-(s)$

There is a more general theory of acids and bases that can be used in more situations:

- *An acid is a proton (hydrogen ion) donor.*
- *A base is a proton (hydrogen ion) acceptor.*

This is called the Brønsted–Lowry theory.

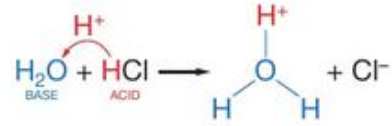


▲ Figure 17.14 A hydrogen ion (H^+) is a proton. If an electron is removed from a hydrogen atom, only a proton is left.

A better way of writing the reaction that occurs when hydrogen chloride dissolves in water is:

$$H_2O(l) + HCl(aq) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

The HCl donates (gives) a proton (H^+) to an H_2O molecule. So, according to our definition, the HCl is an acid and H_2O is a base because it accepts the proton.



▲ Figure 17.15 HCl donates a proton and H_2O accepts it.

The $H_3O^+(aq)$ ion is called the **hydroxonium ion**. This is the ion that we usually write simply as $H^+(aq)$. You can think of it as a hydrogen ion joined to a water molecule. Cl^- has a negative charge because the H^+ left an electron behind when it transferred to the H_2O.

THE REACTION BETWEEN NH_3 AND HCl

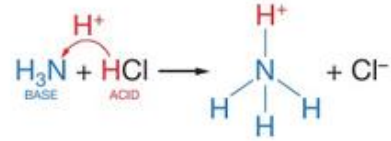
Dilute hydrochloric acid reacts with ammonia solution in a neutralisation reaction to form a salt, ammonium chloride:

$$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$$

Ammonium chloride is an ionic salt so we can show the ions that are present:

$$NH_3(aq) + HCl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

We can see now that, in order to form ammonium chloride, the HCl must donate a proton (H^+) to the NH_3. The HCl is an acid because it donates a proton and the NH_3 is a base because it accepts the proton.



▲ Figure 17.16 HCl is an acid and donates a proton to NH_3 (a base).

MORE ACID-BASE REACTIONS

Acids in solution are acidic because they donate a proton (H^+) to water to form the hydroxonium ion. If we take a general acid (HA), this is what happens when it is put into water:

$$HA(aq) + H_2O(l) \rightarrow H_3O^+(aq) + A^-(aq)$$

HA donates a proton to H_2O, which acts as a base. For nitric acid this would be:

$$\underset{\text{acid}}{HNO_3(aq)} + \underset{\text{base}}{H_2O(l)} \rightarrow H_3O^+(aq) + NO_3^-(aq)$$

When bases, such as metal oxides (e.g. Na_2O), react with water they accept a proton from water, for example:

$$(Na^+)_2O^{2-}(s) + H_2O(l) \rightarrow 2Na^+(aq) + 2OH^-(aq)$$

The O^{2-} ion accepts a proton from the water. In this reaction the water acts as an acid because it donates a proton.

When sulfuric acid reacts with copper(II) oxide (CuO) we have:

$$Cu^{2+}O^{2-}(s) + H_2SO_4(aq) \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq) + H_2O(l)$$

H_2SO_4 is an acid, it donates protons to CuO, the base.

Ammonia is also a base:

$$\underset{\text{base}}{NH_3(aq)} + \underset{\text{acid}}{H_2O(l)} \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

The ammonia accepts a proton (H^+) from the water.

For reactions such as:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

What actually happens is that the H_3O^+ ion formed when the acid reacts with water donates a proton to the base, OH^-.

$$\underset{\text{acid}}{H_3O^+(aq)} + \underset{\text{base}}{OH^-(aq)} \rightarrow 2H_2O(l)$$

But for International GCSE purposes, we almost always use the simplified version:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

HINT

Water can act as proton acceptor (base) or proton donor (acid) in reactions on this page. Do not get confused with what is happening here; remember, water is still neutral! We are simply understanding the reactions that occur in terms of proton transfer.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 5

SKILLS PROBLEM SOLVING 9

1 State which of the following will react with dilute sulfuric acid:

a copper, copper(II) oxide, copper(II) hydroxide, copper(II) carbonate

b In the case of each of the substances which does react, write the equation (including state symbols) for the reaction. All of these substances are insoluble solids.

SKILLS CRITICAL THINKING 7

2 Read this description of the chemistry of metal **A** and some of its compounds, and then answer the questions.

Metal **A** has no reaction with dilute hydrochloric acid or dilute sulfuric acid. It forms a black oxide, **B**, which reacts with hot dilute sulfuric acid to give a blue solution, **C**. Metal **A** also forms a green compound, **D**, which reacts with dilute nitric acid to give a colourless gas, **E**, and another blue solution, **F**. The colourless gas, **E**, turned limewater milky.

a State the names of **A**, **B**, **C**, **D**, **E** and **F**.

SKILLS PROBLEM SOLVING 9

b Write equations for the reactions between:

i **B** and dilute sulfuric acid

ii **D** and dilute nitric acid.

SKILLS CRITICAL THINKING 6

3 a Nickel, Ni, is a silvery metal just above hydrogen in the reactivity series. Nickel(II) compounds in solution are green.

i Describe what you would see if you warmed some nickel with dilute sulfuric acid in a test-tube.

ii State the name of any gas formed.

SKILLS PROBLEM SOLVING 9

iii Write the equation for the reaction between nickel and dilute sulfuric acid.

b Nickel(II) carbonate is a green, insoluble powder. A spatula measure of nickel(II) carbonate is added to some dilute hydrochloric acid in a test-tube.

SKILLS CRITICAL THINKING 6

i Describe what you would observe in this experiment.

SKILLS PROBLEM SOLVING 9

ii Write an equation for the reaction that occurs.

SKILLS EXTENSTION QUESTION 8

iii Write an ionic equation for the reaction that occurs.

SKILLS CRITICAL THINKING 6

4 Organise the following compounds into two lists: those that are soluble in water and those that are insoluble.

sodium chloride, lead(II) sulfate, zinc nitrate, calcium carbonate, iron(III) sulfate, lead(II) chloride, potassium sulfate, copper(II) carbonate, silver chloride, aluminium nitrate, barium sulfate, ammonium chloride, magnesium nitrate, calcium sulfate, sodium phosphate, nickel(II) carbonate, chromium(III) hydroxide, potassium dichromate(VI)

SKILLS EXECUTIVE FUNCTION

5 a Describe in detail the preparation of a pure, dry sample of copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$, starting from copper(II) oxide.

SKILLS PROBLEM SOLVING 9

b Write equations for **i** the reaction producing copper(II) sulfate solution and **ii** the crystallisation reaction.

CHEMISTRY ONLY

SKILLS EXECUTIVE FUNCTION 7

6 a Read the following description of a method for making sodium sulfate crystals, $Na_2SO_4 \cdot 10H_2O$, and then explain the reasons for each of the underlined phrases or sentences.

25 cm^3 of sodium carbonate solution was transferred to a conical flask using a pipette, and a few drops of methyl orange were added. Dilute sulfuric acid was run in from a burette until the solution became orange. The volume of acid added was noted. That same volume of dilute sulfuric acid was added to a fresh 25 cm^3 sample of sodium carbonate solution in a clean flask, but without the methyl orange. The mixture was evaporated until a sample taken on the end of a glass rod crystallised on cooling in the air. The solution was left to cool. The crystals formed were separated from the remaining solution and dried.

SKILLS PROBLEM SOLVING

b Write equations for i the reaction producing sodium sulfate solution and ii the crystallisation reaction.

SKILLS REASONING, PROBLEM SOLVING

7 Suggest solutions that could be mixed together to make each of the following insoluble salts. In each case write the equation for the reaction you choose.

a Silver chloride
b Calcium carbonate
c Lead(II) sulfate
d Lead(II) chloride

SKILLS EXECUTIVE FUNCTION, PROBLEM SOLVING

8 Describe in detail the preparation of a pure, dry sample of barium carbonate. Write the equation for the reaction you use.

9 There are three main methods of making salts:

A Reacting an acid with an excess of a suitable solid.

B Using a titration.

C Using a precipitation reaction.

For each of the following salts, write down the letter of the appropriate method, and state the names of the substances you would react together. You should state whether they are used as solids or solutions. Write an equation for each reaction.

a Zinc sulfate
b Barium sulfate
c Potassium nitrate
d Copper(II) nitrate
e Lead(II) chromate(VI) (a bright yellow insoluble solid; chromate(VI) ions have the formula CrO_4^{2-}).

END OF CHEMISTRY ONLY

SKILLS REASONING

10 All of the following equations represent acid–base reactions. For each of the equations, state which substance is the acid and which the base.

a $CO_3^{2-}(s) + H_3O^+(aq) \rightarrow HCO_3^-(aq) + H_2O(l)$
b $MgO(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2O(l)$
c $HNO_3(aq) + NH_3(aq) \rightarrow NH_4^+(aq) + NO_3^-(aq)$
d $H_2SO_4(aq) + H_2O(l) \rightarrow HSO_4^-(aq) + H_3O^+(aq)$
e $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$
f $CH_3COOH(aq) + NH_3(aq) \rightarrow CH_3COO^-(aq) + NH_4^+(aq)$